

# Zaher Karp, MPH

me@zaherkarp.com | zaherkarp.com | linkedin.com/in/zkarp

## Summary

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Data platform leader and health services researcher with 15+ years spanning academic research and production analytics engineering in regulated healthcare. Deep expertise in HEDIS, CMS Medicare Stars, and HIPAA-compliant data handling across Epic, Cerner, Veradigm, and athenahealth, with a peer-reviewed publication record in primary care systems engineering, EHR optimization, and accountable care.

## Experience

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Baltimore Health Analytics | Manager of Data Science & Engineering    Nov 2025 – Present

MariaDB, Python, JavaScript, Ruby on Rails, Docker, Perl

- Own the data platform for a Medicare Advantage analytics product, including ETL/ELT architecture, stored procedure design, and the regulatory validation layer against CMS technical specifications
- Hardened and validated an ETS forecasting pipeline for Stars cutpoint prediction under regulatory scrutiny, reducing forecast error 34–53% versus standard Holt-Winters by downweighting COVID-era outliers with robust M-estimators
- Built HEDIS TRC, BPD, and CBP benchmarking pipelines using a pooled top-decile denominator methodology to produce structurally defensible performance ceilings
- Ported a legacy Perl pharmacy claims pipeline (Diabetes, RAS Antagonists, Statins PDC) to Python with pre-flight validation, structured logging, and MariaDB compatibility
- Lead a distributed engineering and QA team across schema design, code review, and release decisions; design data quality frameworks and release-readiness rubrics

Healthfinch (acquired by Health Catalyst) | Analytics Manager    Dec 2017 – Aug 2025

AWS (S3, Redshift), Azure, Databricks, dbt, SQL, Python (pandas, PySpark, Jinja), Airflow, Tableau, Sisense, DataDog

- First analytics hire; built HIPAA-certified data infrastructure on AWS, promoted to Manager within a year

- Built and scaled the data platform supporting value-based care and billing analytics across 50+ health systems
- Led analytics integration through the Health Catalyst acquisition, migrating infrastructure from AWS to Azure and Databricks and rebuilding the analytics layer using medallion architecture (bronze/silver/gold) while maintaining continuity of service across the combined portfolio
- Reduced AWS S3 storage costs ~50% and ETL load time 24+ hours via storage-tier redesign and partitioning
- Standardized legacy analytics into governed ELT pipelines (~70% codebase reduction); established semantic layers and data governance practices enabling 7x Sisense user growth and eliminating 400+ manual hours per quarter
- Designed HIPAA and HITRUST compliant security architecture and observability for analytics platforms; built ROI models supporting \$1MM+ in recurring revenue decisions

University of Wisconsin-Madison | Researcher and Research Specialist Sep 2009 – Jun 2018

SAS, Stata, R, NVivo, Qualtrics, Epic

- Co-authored 6+ peer-reviewed publications on accountable care shared savings, cancer screening performance, and EHR optimization across Family Medicine, Internal Medicine, Pediatrics, and Population Health Sciences
- Supported \$250,000+ in federal grants (AHRQ, NIH, PCORI) through quantitative and qualitative analysis
- Designed and executed mixed-methods studies on primary care team workflows, environmental design, and the dissemination of practice-level improvements

## *Publications*

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2019 Karp Z, Kamnetz S, Wietfeldt N, Sinsky C, Molfetter T, Pandhi N. Influence of environmental design on team interactions across 3 family medicine clinics. *Health Environments Research & Design Journal* 12(4):159–173. ([PubMed](#) · [ResearchGate](#) · 12 citations)

2019 Pandhi N, Gaines ME, Deci D, Schlesinger M, Culp C, Karp Z et al. Broadening medical students' exposure to the range of illness experiences: a pilot experimental curriculum trial. *Academic Medicine*. ([PubMed](#) · 4 citations)

2018 Berkson S, Davis S, Karp Z, Jaffery J, Flood G, Pandhi N. Medicare Shared Savings Programs: higher cost accountable care organizations are more likely to achieve savings. *International Journal of Healthcare Management*.

- 2016 Pandhi N, Jacobson N, Serrano N, Hernandez A, Zeidler-Schreiter E, Wietfeldt N, Karp Z. An efficient process of gathering diverse community opinions to inform an intervention. *Implementation Science* 11(Suppl 1):A13. ([Full text](#))
- 2014 Pandhi N, Yang WL, Karp Z, Young A, Beasley JW, Kraft S, Carayon P. Approaches and challenges to optimizing primary care teams' electronic health record usage. *Journal of Innovation in Health Informatics* 21(3):142-51. ([PubMed](#) · 24 citations)
- 2012 Yang W, Pandhi N, Karp Z, Young A, Beasley J, Kraft S, Carayon P. Approaches and challenges to optimizing the use of EHRs in primary care (preliminary findings). *Proceedings of World Conference on E-Learning, Montréal*. ([Full text](#))

## *Presentations*

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Seventeen podium presentations, workshops, and posters at national and regional venues between 2010 and 2017, in healthcare services research and primary care systems engineering. A co-authored poster received the Patient Choice Award (2 of 45) at the North American Primary Care Research Group Conference, 2015.

- 2017 National Collaborative for Improving Primary Care Through Industrial and Systems Engineering. Madison, WI.
- 2017 UW Institute for Clinical and Translational Research, Dissemination and Implementation Short Course. Madison, WI.
- 2016 AAMC Integrating Quality Meeting. Chicago, IL.
- 2016 National Collaborative for Improving Primary Care Through Industrial and Systems Engineering. Madison, WI.
- 2015 Field Innovation Team's Bootcamp. Provo, UT.
- 2015 Access, Quality, and Outcomes Research Network. Appleton, WI.
- 2015 Society for Implementation Research Collaboration Conference. Seattle, WA.
- 2015 National Collaborative for Improving Primary Care Through Industrial and Systems Engineering. Madison, WI.
- 2015 UW Health Quality Week. Madison, WI.
- 2015 North American Primary Care Research Group Conference. Cancún, Mexico.
- 2015 Wisconsin Research and Education Network Convocation of Practices. Oshkosh, WI.

2014 National Collaborative for Improving Primary Care Through Industrial and Systems Engineering. Madison, WI.

2014 AAMC Integrating Quality Meeting. Chicago, IL.

2014 Pharmacy Society of Wisconsin Annual Meeting. Madison, WI.

2012 World Conference on E-Learning in Corporate, Government, Healthcare, and Higher Education. Montréal, QC.

2012 UW Health organizational leadership. Middleton, WI.

2010 Wisconsin Primary Care Research and Quality Improvement Forum. Middleton, WI.

### *Grants and Funding*

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Contributed quantitative and qualitative analysis to more than \$250,000 in federal research funding from the Agency for Healthcare Research and Quality (AHRQ), the National Institutes of Health (NIH), and the Patient-Centered Outcomes Research Institute (PCORI), supporting studies of primary care redesign, EHR optimization, and accountable care performance.

### *Education*

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Master of Public Health, Biostatistics | University of Wisconsin-Madison | Dec 2015

Graduate Certificate, Patient Safety (Industrial & Systems Engineering) | University of Wisconsin-Madison | Dec 2015

Bachelor of Arts, English Literature | University of Wisconsin-Madison | May 2007

### *Service and Professional Activities*

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Peer reviewer for healthcare services and informatics journals, 2019 to present

Board member, The Road Home Dane County, 2016 to 2020

Undergraduate research mentor, University of Wisconsin-Madison, 2014 to 2017

Board chair, WORT 89.9 FM community radio, 2013 to 2015

Lean Six Sigma certification, 2015

## *Awards and Honors*

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Alumni Hall Service Award, University of Wisconsin–Madison, 2017

Patient Choice Award (2 of 45), North American Primary Care Research Group Conference, 2015

## *Skills*

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Data platform: SQL (Postgres, SQL Server, MySQL, MariaDB), Python (pandas, PySpark), dbt, Airflow, Docker, AWS, Azure, Databricks, Tableau, Sisense, DataDog, R, git

Healthcare domain: HEDIS, CMS Medicare Stars, HIPAA, HITRUST, EHR systems (Epic Clarity, Cerner, Veradigm, athenahealth), FHIR, HL7, CMS technical specifications, value-based care, ACO / MSSP

Research methods: biostatistics, interrupted time series, mixed-methods study design, qualitative coding (NVivo), survey instrumentation (Qualtrics), SAS, Stata

Governance and delivery: ETL/ELT architecture, data modeling, semantic layers, metric governance, release readiness, data quality frameworks, cross-functional stakeholder communication